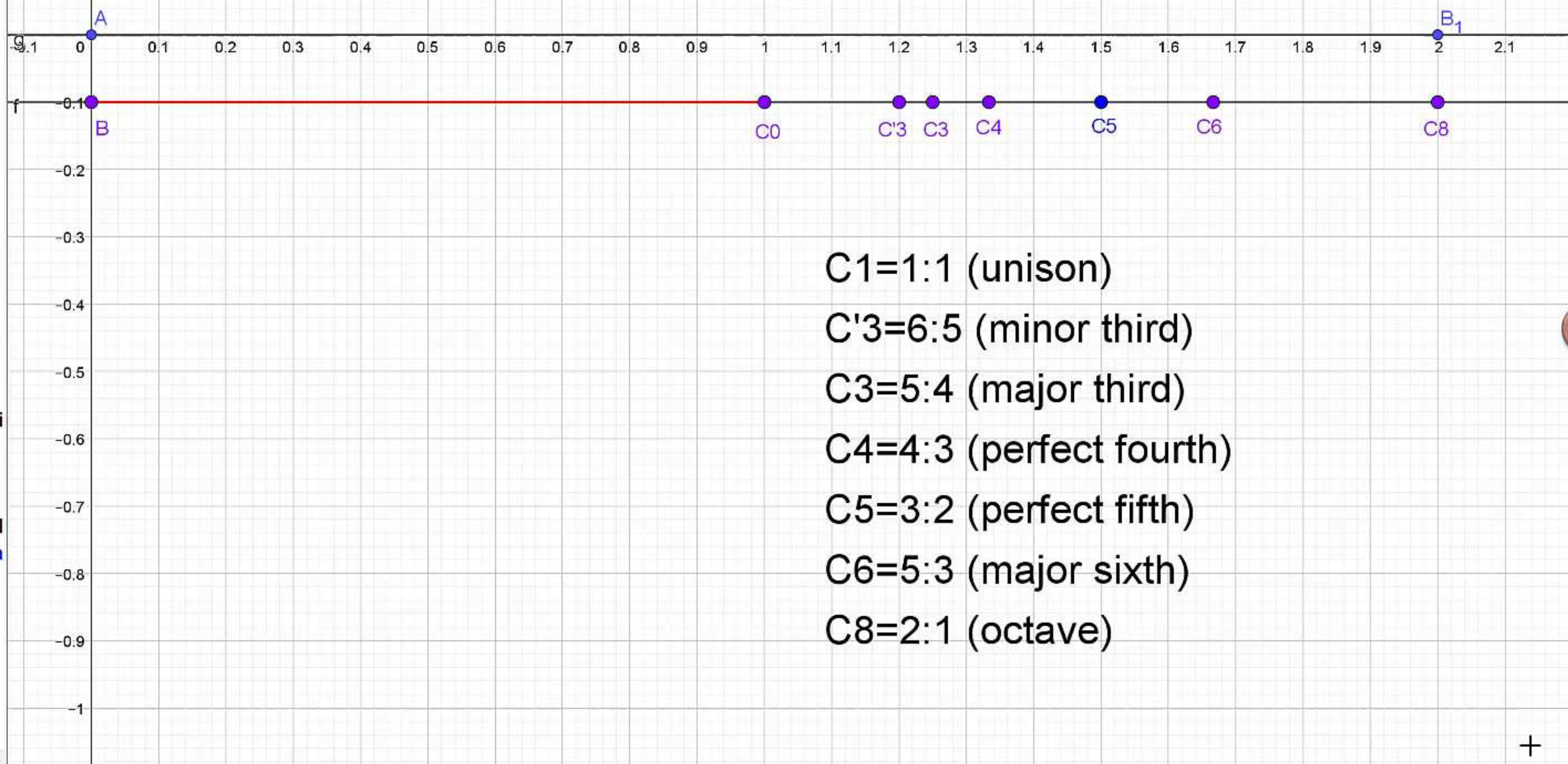


Algebra

Graphics

● $A = (0, 0)$
 ○ $N0 = (1, 0)$
 ○ $N2 = (1.12246, 0)$
 ○ $N3 = (1.18921, 0)$
 ○ $N4 = (1.25992, 0)$
 ○ $N5 = (1.33484, 0)$
 ○ $N6 = (1.41421, 0)$
 ○ $N7 = (1.49831, 0)$
 ○ $N8 = (1.5874, 0)$
 ○ $N9 = (1.68179, 0)$
 ○ $N10 = (1.7818, 0)$
 ○ $N11 = (1.88775, 0)$
 ○ $N1 = (1.05946, 0)$
 ● $B_1 = (2, 0)$
 ● $C5 = (1.5, -0.1)$
 ● $f: y = -0.1$
 ● $g: y = 0$
 ● $C4 = (1.33333, -0.1)$
 ● $C3 = (1.25, -0.1)$
 ● $C'3 = (1.2, -0.1)$
 ● $C6 = (1.66667, -0.1)$
 ● $C0 = (1, -0.1)$
 ● $C8 = (2, -0.1)$
 ● $\text{text1} = "C1=1:1 (\text{uni}$
 ● $B = (0, -0.1)$
 ○ $C2 = (1.125, -0.1)$
 ○ $C1 = (1.05882, -0.1)$
 ○ $\text{text2} = "C2=9:8 (\text{full}$
 ● $\text{text3} = "Pythagorea$
 ● $h = 1$

Pythagorean Intervals and Beyond



$C1=1:1$ (unison)

$C'3=6:5$ (minor third)

$C3=5:4$ (major third)

$C4=4:3$ (perfect fourth)

$C5=3:2$ (perfect fifth)

$C6=5:3$ (major sixth)

$C8=2:1$ (octave)

Input:



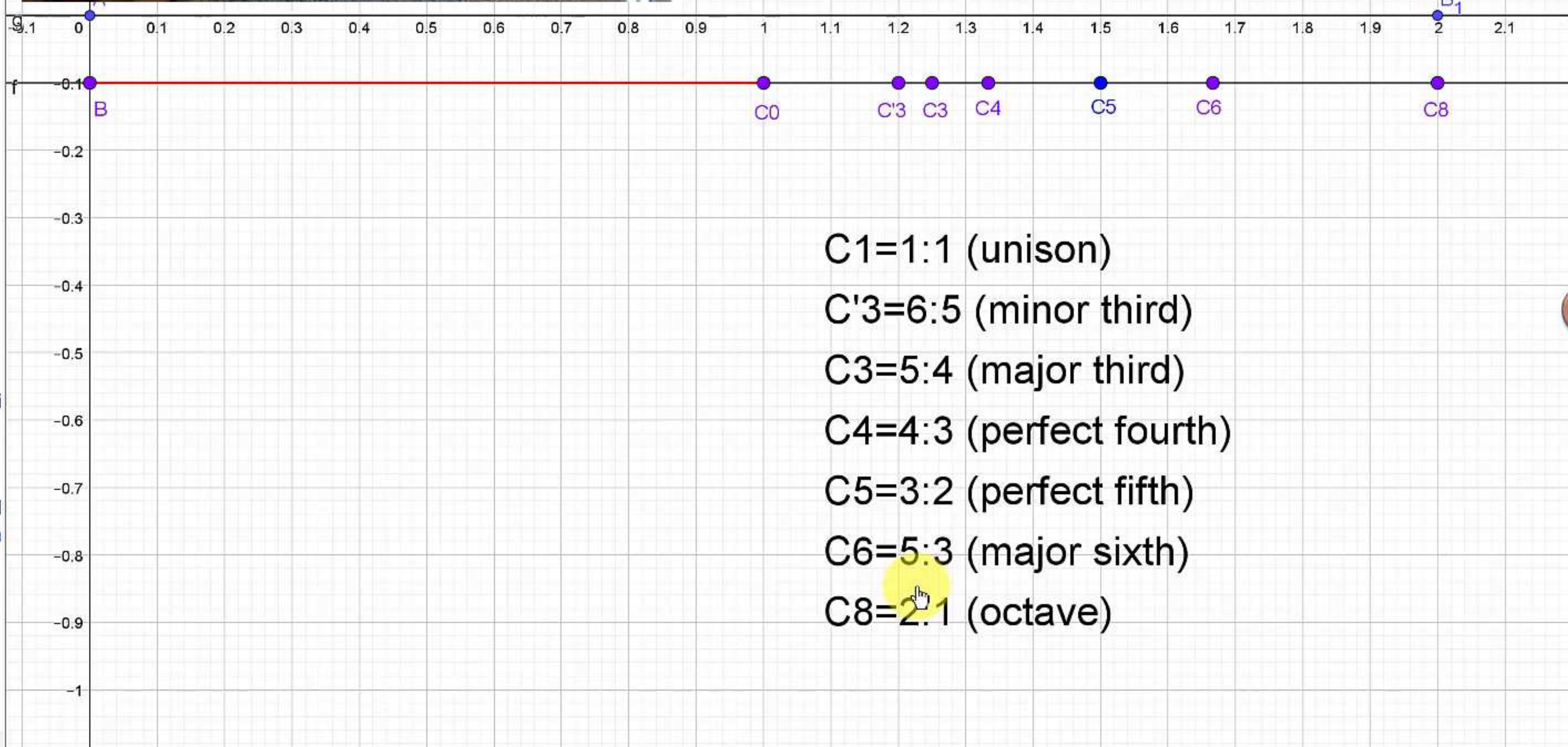


Algebra

- $A = (0, 0)$
- $N0 = (1, 0)$
- $N2 = (1.12246, 0)$
- $N3 = (1.18921, 0)$
- $N4 = (1.25992, 0)$
- $N5 = (1.33484, 0)$
- $N6 = (1.41421, 0)$
- $N7 = (1.49831, 0)$
- $N8 = (1.5874, 0)$
- $N9 = (1.68179, 0)$
- $N10 = (1.7818, 0)$
- $N11 = (1.88775, 0)$
- $N1 = (1.05946, 0)$
- $B_1 = (2, 0)$
- $C5 = (1.5, -0.1)$
- $f: y = -0.1$
- $g: y = 0$
- $C4 = (1.33333, -0.1)$
- $C3 = (1.25, -0.1)$
- $C'3 = (1.2, -0.1)$
- $C6 = (1.66667, -0.1)$
- $C0 = (1, -0.1)$
- $C8 = (2, -0.1)$
- $\text{text1} = "C1=1:1 (\text{uni})"$
- $B = (0, -0.1)$
- $C2 = (1.125, -0.1)$
- $C1 = (1.05882, -0.1)$
- $\text{text2} = "C2=9:8 (\text{full})"$
- $\text{text3} = "Pythagorea"$
- $h = 1$



Beyond



$C1=1:1$ (unison)

$C'3=6:5$ (minor third)

$C3=5:4$ (major third)

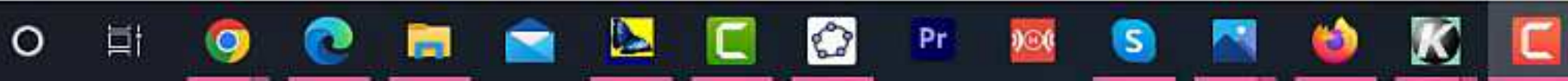
$C4=4:3$ (perfect fourth)

$C5=3:2$ (perfect fifth)

$C6=5:3$ (major sixth)

$C8=2:1$ (octave)

Input:

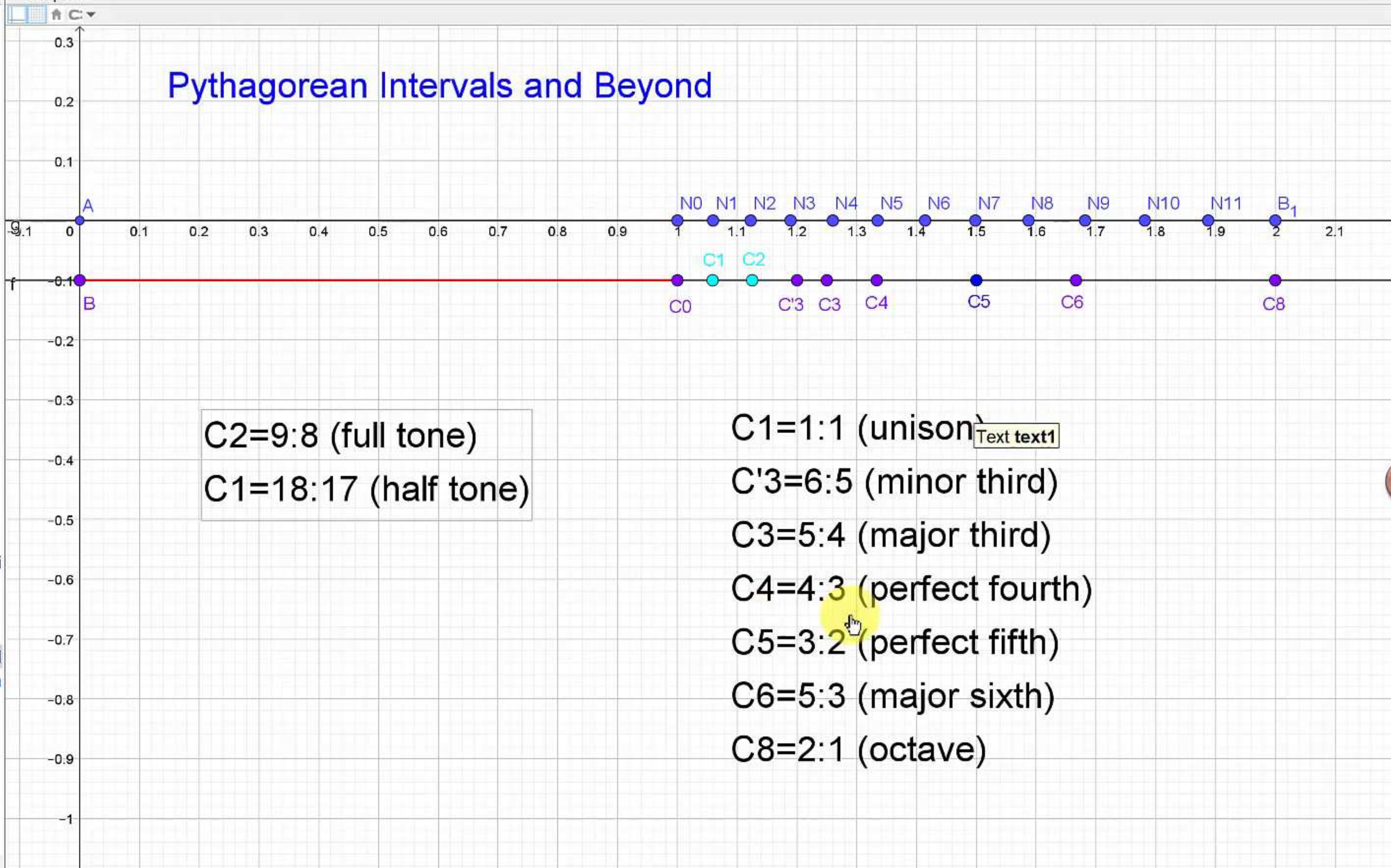




Algebra

- $A = (0, 0)$
- $N0 = (1, 0)$
- $N2 = (1.12246, 0)$
- $N3 = (1.18921, 0)$
- $N4 = (1.25992, 0)$
- $N5 = (1.33484, 0)$
- $N6 = (1.41421, 0)$
- $N7 = (1.49831, 0)$
- $N8 = (1.5874, 0)$
- $N9 = (1.68179, 0)$
- $N10 = (1.7818, 0)$
- $N11 = (1.88775, 0)$
- $N1 = (1.05946, 0)$
- $B_1 = (2, 0)$
- $C5 = (1.5, -0.1)$
- $f: y = -0.1$
- $g: y = 0$
- $C4 = (1.33333, -0.1)$
- $C3 = (1.25, -0.1)$
- $C'3 = (1.2, -0.1)$
- $C6 = (1.66667, -0.1)$
- $C0 = (1, -0.1)$
- $C8 = (2, -0.1)$
- $\text{text1} = "C1=1:1 (\text{uni}$
- $B = (0, -0.1)$
- $C2 = (1.125, -0.1)$
- $C1 = (1.05882, -0.1)$
- $\text{text2} = "C2=9:8 (\text{full}$
- $\text{text3} = "Pythagorea$
- $h = 1$
- $a = 1.05946$

Graphics



Input:

